***Solution:***

Computation of three- yearly moving averages.

| Year | Number of students | 3- yearly moving Total | 3- yearly moving Averages |
|---|---|---|---|
| 1995 | 332 | --- | --- |
| 1996 | 317 | 1006 | 335.33 |
| 1997 | 357 | 1066 | 355.33 |
| 1998 | 392 | 1151 | 383.67 |
| 1999 | 402 | 1199 | 399.67 |
| 2000 | 405 | 1217 | 405.67 |
| 2001 | 410 | 1242 | 414.00 |
| 2002 | 427 | 1272 | 424.00 |
| 2003 | 435 | 1300 | 433.33 |
| 2004 | 438 | --- | --- |

Table 9.3

**Example 9.5**

Calculate four-yearly moving averages of number of students studying in a higher secondary school in a particular city from the following data.

| Year | Number of students |
|---|---|
| 2001 | 124 |
| 2002 | 120 |
| 2003 | 135 |
| 2004 | 140 |
| 2005 | 145 |
| 2006 | 158 |
| 2007 | 162 |
| 2008 | 170 |
| 2009 | 175 |

***Solution:***

Computation of four- yearly moving averages.

| Year | Sales | 4-yearly centered moving total | 4-yearly moving Average | 4-yearly centered moving Average |
|---|---|---|---|---|
| 2001 | 124 | -- | -- | -- |
| | | | | |
| 2002 | 120 | -- | -- | -- |
| | | 519 | 129.75 | |
| 2003 | 135 | -- | | 132.37 |
| | | 540 | 135.00 | |
| 2004 | 140 | -- | | 139.75 |
| | | 578 | 144.50 | |
| 2005 | 145 | -- | | 147.87 |
| | | 605 | 151.25 | |
| 2006 | 158 | -- | | 155.00 |
| | | 635 | 158.75 | |
| 2007 | 162 | -- | | 162.50 |
| | | 665 | 166.25 | |
| 2008 | 170 | -- | -- | - |
| | | | | |
| 2009 | 175 | -- | -- | - |

Table 9.4

**Note**

The calculated 4-yearly centered moving average belongs to the particular year present in that row. In example 9.5, the value 132.37 belongs to the year 2003.

## 9.1.4 Method of Least Squares

The line of best fit is a line from which the sum of the deviations of various points is zero. This is the best method for obtaining the trend values. It gives a convenient basis for calculating the line of best fit for the time series. It is a mathematical method for measuring trend. Further the sum of the squares of these deviations would be least when compared with other fitting methods. So, this method is known as the Method of Least Squares and satisfies the following conditions:

(i) The sum of the deviations of the actual values of $Y$ and $\hat{Y}$ (estimated value of $Y$) is Zero. that is $\Sigma(Y-\hat{Y}) = 0$.

(ii) The sum of squares of the deviations of the actual values of $Y$ and $\hat{Y}$ (estimated value of $Y$) is least. that is $\Sigma(Y-\hat{Y})^2$ is least.



XII Std - Business Maths & Stat EM Chapter 9.indd 205 2/28/2020 5:11:03 PM